

1 There are two production workshops and an assembly workshop in a factory. Two production workshops produce part A and B respectively. The task of assembly workshop is to assemble A and B into a product. Part A should be sent to the shop shelves F1 and part B should be sent to the shop shelves F2. Both of shop shelves can store 10 parts at most. The assembler assembles into a product by a part A and a part B from the shelves every time.

Please design semaphore-based programs to describe their activities. It is required:

- 1) To define the semaphores, explain the role of each semaphore, and give their initial values;
- 2) To illustrate the structures of programs for two production workshops and an assembly workshop.

2

In a page system, the page table is stored in main memory.

- (1) If it takes 5us to access main memory, how long does it cost for accessing to a page?
- (2) If there is a TLB, and the hit ratio is 80%. Ignore the time for searching page table entry in TLB, how long does it cost for accessing to a page?

3

某计算机主存按字节编址，逻辑地址和物理地址都是 32 位，页表项大小为 4 字节。请回答下列问题。

若使用一级页表的分页存储管理方式，逻辑地址结构为：

页号（20 位） 页内偏移量（12 位）

则页的大小是多少字节？

页表最大占用多少字节？